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Project Details

Project Title:	AI-based Threat Detection & Elimination for Ransomware/Zero-day Attacks	
Type of Project	AI-based cybersecurity prototype Project	
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Abstract:

The rapid advancement of digital technologies has significantly increased the dependence of individuals, organizations, and governments on interconnected systems. While this digital transformation has enabled efficiency and innovation, it has also led to a sharp rise in sophisticated cyber threats. Among these, ransomware and zero-day attacks have emerged as two of the most destructive and difficult-to-detect forms of cyberattacks. Traditional security solutions that rely on signature-based detection are ineffective against such evolving threats, as they fail to identify previously unseen attack patterns.

This project presents an AI-based threat detection and elimination system that uses behavior-based machine learning techniques to identify ransomware and zero-day attacks. The proposed system analyzes system-level and file-level behavioral features such as file entropy, frequency of file modifications, CPU utilization, memory usage, and network activity. A Random Forest classifier is trained on these features to distinguish between benign and malicious behavior.

Once a threat is detected, the system automatically initiates a response by isolating the suspicious file in a quarantine environment and recording the incident in security logs. The project demonstrates a functional prototype of an intelligent endpoint security system that emphasizes proactive threat detection, automated mitigation, and extensibility for future enhancements. This work validates the feasibility of AI-driven cybersecurity solutions in addressing modern ransomware and zero-day attack challenges.

1. Introduction:

Cybersecurity has become one of the most critical concerns in the modern digital era. The increasing adoption of cloud computing, Internet of Things (IoT), remote work environments, and digital services has significantly expanded the attack surface for cybercriminals. As a result, cyberattacks have become more frequent, sophisticated, and damaging.

Among the various forms of cyber threats, **ransomware** and **zero-day attacks** pose the greatest risk. Ransomware attacks encrypt sensitive data and demand ransom payments, often leading to operational shutdowns, financial losses, and reputational damage. Zero-day attacks exploit unknown vulnerabilities in software systems before developers or security vendors become aware of them, making them extremely difficult to detect and prevent.

Traditional cybersecurity mechanisms rely primarily on **signature-based detection**, where known malware patterns are matched against incoming threats. While effective against known attacks, these systems fail when faced with novel or modified malware variants. Modern attackers frequently use polymorphic malware, fileless attacks, and obfuscation techniques to bypass signature-based defenses.

Artificial Intelligence (AI) and Machine Learning (ML) have emerged as promising solutions to overcome these limitations. AI-based security systems focus on **behavioral analysis**, anomaly detection, and pattern recognition, enabling the detection of previously unseen threats. By learning from historical data and adapting to new patterns, AI-driven systems provide a proactive and scalable approach to cybersecurity.

This project aims to leverage AI techniques to design and implement a threat detection and elimination system capable of identifying ransomware and zero-day attacks based on behavioral characteristics rather than predefined signatures.

In this context, behavior-based threat detection has gained significant importance, as it focuses on monitoring how programs interact with system resources rather than relying on static malware signatures. Ransomware typically exhibits distinct behavioral patterns such as rapid file modifications, high entropy changes due to encryption, abnormal CPU and memory usage, and unusual network communication. By analyzing these indicators using machine learning models, it becomes possible to identify malicious activities at an early stage, even when the attack is previously unknown. The proposed system adopts this behavior-driven approach to improve detection accuracy and response speed, thereby minimizing potential damage and strengthening overall system security.

Furthermore, the use of machine learning models in cybersecurity offers adaptability and scalability, which are essential in today's dynamic threat landscape. Unlike rule-based systems that require constant manual updates, machine learning models can be retrained with new data to improve detection accuracy over time. This adaptability allows security solutions to evolve alongside emerging attack techniques. The proposed project demonstrates how a lightweight machine learning model can be integrated into a practical threat detection framework, laying the foundation for more advanced and real-time security solutions in future developments.

2. Literature Survey:

The growing complexity of cyber threats has motivated researchers to explore Artificial Intelligence (AI) and Machine Learning (ML) techniques for detecting ransomware and zero-day attacks. Traditional signature-based security mechanisms are ineffective against unknown and evolving threats, leading to increased interest in behavior-based and learning-driven approaches.

In recent years, deep learning techniques have gained prominence in zero-day attack detection. In 2025, a significant contribution was made through the work "Explainable AI for Zero-Day Attack Detection in IoT Networks Using Attention Fusion Model" [1]. This study proposed an attention-fusion deep learning architecture combined with Explainable AI (XAI) techniques to detect previously unseen zero-day attacks in IoT environments. The use of attention mechanisms enabled the model to focus on the most relevant features, while XAI improved transparency and trust in AI-driven security systems.

Similarly, Nandiraju et al. [2] presented a deep learning-based approach for detecting zero-day attacks in network traffic. Their work, published in the Asian Journal of Research in Computer Science in 2025, utilized a Convolutional Neural Network (CNN) trained on the UNSW-NB15 dataset. The results demonstrated high detection accuracy for unknown attacks, highlighting the effectiveness of deep learning models in identifying complex attack patterns within large-scale network data.

While deep learning models provide strong detection capabilities, their deployment in real-time environments often faces challenges related to computational overhead. Addressing this issue, Arjun Sekar et al. [3] proposed an innovative framework that integrates extended Berkeley Packet Filter (eBPF) based kernel tracing with AI and Natural Language Processing (NLP) techniques. Their 2024 study demonstrated real-time ransomware detection by analyzing low-level system behavior, offering a practical approach for endpoint protection systems.

Machine learning classifiers have also shown promising results in ransomware and zero-day detection. In 2024, Nhlapo and Nkongolo [4] conducted a comparative evaluation of multiple ML models, including Random Forest and XGBoost, for detecting ransomware and zero-day attacks. Their findings indicated that ensemble-based models such as Random Forest provide high accuracy, robustness, and resistance to overfitting, making them suitable for behavior-based threat detection systems.

To provide a broader perspective, Yang Guo [5] presented a comprehensive survey in Computer Communications (2023), reviewing various machine learning techniques used for zero-day attack detection. The study discussed challenges such as data imbalance, feature selection, model interpretability, and scalability. The survey emphasized the importance of behavior-based detection and highlighted Random Forest and hybrid ML approaches as effective solutions for identifying unknown threats.

From the literature, it is evident that AI-based behavioral analysis is a powerful approach for detecting ransomware and zero-day attacks. While deep learning models offer high accuracy, traditional machine learning classifiers such as Random Forest provide a balanced trade-off between performance, interpretability, and computational efficiency. These insights directly influenced the design of the proposed project, which adopts a behavior-based machine learning approach for effective threat detection and automated mitigation.

3. Positioning

3.1. Problem statement: The modern digital ecosystem is highly vulnerable to ransomware and zero-day attacks due to the limitations of traditional cybersecurity solutions. Existing security systems are unable to detect unknown threats in real time, resulting in delayed response and extensive damage. There is a critical need for an intelligent, automated system that can detect and mitigate ransomware and zero-day attacks using adaptive and behavior-based techniques.

3.2. Product position statement: The proposed system is positioned as an AI-driven, behavior-based endpoint security prototype designed to detect and eliminate ransomware and zero-day threats. Unlike conventional antivirus solutions, this system focuses on analyzing system behavior using machine learning to identify malicious activities and respond automatically through quarantine and logging mechanisms.

4. Planning

Stakeholder Descriptions

Project stakeholders are individuals or groups that have an interest in the project or are affected by its development, execution, and outcomes. In the context of this project, stakeholders include users, organizations, and technical teams who benefit directly or indirectly from an AI-based threat detection system. These stakeholders play an important role in defining requirements, validating outcomes, and ensuring the relevance of the project.

The successful implementation of an AI-based ransomware and zero-day attack detection system contributes to improved cybersecurity awareness, enhanced protection of digital assets, and reduced risk of cyber incidents for all involved stakeholders.

4.1. User stakeholders:

User stakeholders are the primary beneficiaries of this project. They interact with the system either directly or indirectly and benefit from its outputs.

The key user stakeholders for this project include:

- **Organizations and Enterprises** that require protection for systems, files, and sensitive data from ransomware attacks.
- **Small and Medium Businesses (SMBs)** that lack advanced cybersecurity infrastructure but are highly vulnerable to cyber threats.
- **Cybersecurity Analysts and IT Administrators** who can use the system for

monitoring, analysis, and automated response.

- **Individual Users** seeking protection for personal systems against unknown malware and ransomware threats.
- **Academic and Research Communities** that can use the prototype as a reference for AI-based cybersecurity research.

5. Project Scope

Project overview:

This project focuses on the design and development of an AI-based threat detection and elimination system aimed at identifying ransomware and zero-day attacks using behavior-based machine learning techniques. The system analyzes system behavior patterns and classifies them as benign or malicious using a trained machine learning model.

The project involves defining objectives, assessing feasibility, identifying alternatives, estimating budget, determining key deliverables, and selecting appropriate methodologies and resources. The current phase (Review-2) demonstrates a functional prototype that validates the feasibility of the proposed approach.

5.1. Objectives:

The primary objective of this project is to design and implement an AI-driven system capable of detecting ransomware and zero-day attacks based on behavioral characteristics rather than predefined signatures.

The specific objectives include:

- To analyze system and file behavior for detecting malicious activity
- To train a machine learning model for threat classification
- To implement automated threat elimination through quarantine
- To log detected incidents for monitoring and analysis
- To validate the feasibility of AI-based behavioral detection

5.2. Goals:

The goal of this project is to provide a clear and practical demonstration of how Artificial Intelligence can be applied to modern cybersecurity challenges. The project aims to present a concise and understandable system that showcases detection, response, and logging mechanisms, making it suitable for academic evaluation and future enhancement.

5.3. Feasibility Study:

A feasibility study was conducted to evaluate the practicality of the project from multiple perspectives:

- **Technical Feasibility:**
  The project is technically feasible using Python and open-source machine learning

libraries such as Scikit-learn. The required computational resources are minimal and readily available.

- **Economic Feasibility:**
  The project uses only open-source tools, resulting in negligible cost. No commercial software licenses are required.
- **Operational Feasibility:**
  The system is easy to operate and can be executed on standard personal computers without specialized infrastructure.
- **Schedule Feasibility:**
  The project is achievable within the given academic timeline, including implementation, testing, and documentation.

5.4. Alternatives:

Several alternative approaches were considered during the project planning phase:

- Signature-based antivirus systems
- Rule-based intrusion detection systems
- Deep learning-based threat detection models

Signature-based and rule-based systems were rejected due to their inability to detect unknown threats. Deep learning models were considered but not selected due to higher computational complexity and dataset requirements. A machine learning-based behavioral approach was chosen as the most balanced and feasible solution.

5.5. Budget:

The project budget estimation considers software tools, hardware requirements, and development resources. Since the project relies entirely on open-source software and existing hardware, the overall cost is minimal. The primary resources include a personal computer, development tools, and internet access for research. As a result, the project is highly cost-effective and suitable for academic implementation.

5.6. Key deliverables:

The key deliverables of this project include:

- AI-based threat detection prototype
- Trained machine learning model
- Automated quarantine mechanism
- Security logging system
- Project documentation and reports
- PPT presentation for reviews and final evaluation

5.7. Necessary materials:

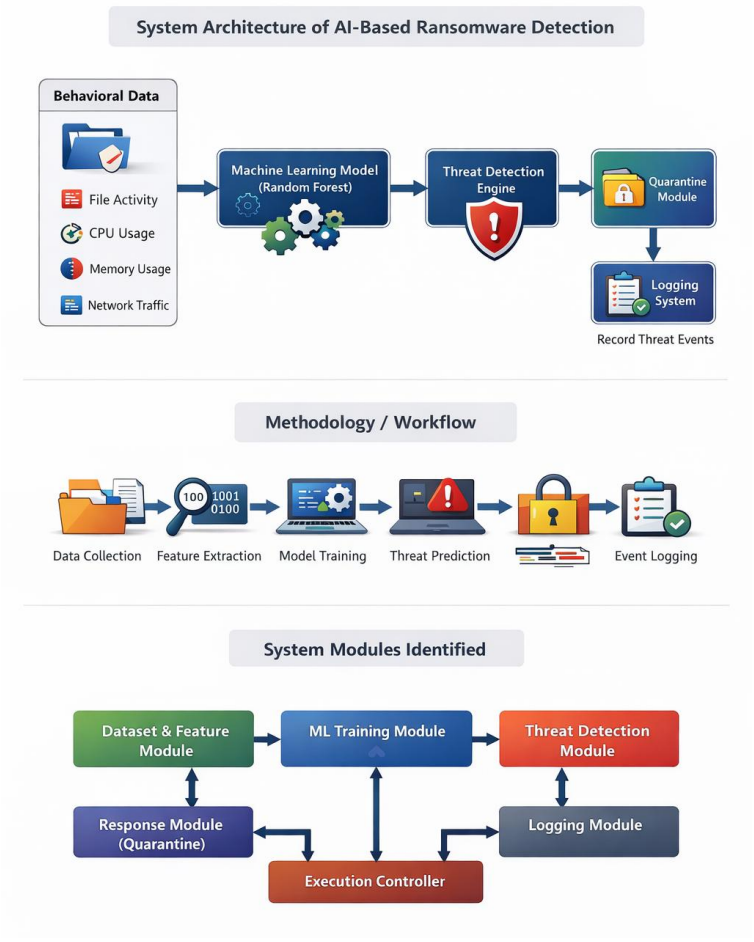
The materials required for this project include:

- Computer system with minimum 8 GB RAM
- Python programming environment
- Open-source machine learning libraries

- Development tools such as Visual Studio Code
- Internet access for research and documentation
- Stationery and digital tools for report preparation

6. Methodology:

The project follows a structured methodology consisting of data collection, feature extraction, machine learning model training, threat detection, automated response, and logging. A modular approach is adopted to ensure clarity, maintainability, and scalability. The methodology emphasizes behavior-based detection to address zero-day and ransomware threats effectively.



7. Modules identified:

The system is divided into the following modules:

- Dataset and Feature Engineering Module
- Machine Learning Training Module
- Threat Detection Module
- Response and Quarantine Module
- Logging and Monitoring Module
- Execution Controller Module

Each module performs a specific function and interacts with other modules to ensure seamless system operation.

8. Requirement and Cost Analysis

Software Requirements

- Python 3.x
- Pandas
- Scikit-learn
- Visual Studio Code

Hardware Requirements

- Minimum 8 GB RAM
- Standard multi-core processor

Cost Analysis

All tools and libraries used are open-source, resulting in zero licensing cost. The overall project cost is minimal, making the solution affordable and scalable.

9. Developing Project Execution Timelines

The project execution followed a structured timeline, including planning, implementation, testing, and review phases. Tasks were distributed across weeks to ensure steady progress and timely completion.

10. Survey Paper Publication Details

A comprehensive literature survey has been completed as part of this project. Research papers from IEEE, Springer, and arXiv were reviewed. A survey paper draft is currently under preparation, and publication is planned during the final SEE phase.

11. Conclusion:

In conclusion, this project demonstrates the feasibility of using Artificial Intelligence for detecting and mitigating ransomware and zero-day attacks. The behavior-based machine learning approach overcomes the limitations of traditional security systems and enables proactive threat detection. The successful implementation of a working prototype in Review-2 establishes a strong foundation for further enhancements in the final phase. The project highlights the effectiveness of AI-driven cybersecurity solutions in addressing modern digital threats.

12. References:

[1] Explainable AI for Zero-Day Attack Detection in IoT Networks Using Attention Fusion Model, Discover Internet of Things, 2025. This paper proposes an attention-fusion deep learning model with XAI techniques for detecting new zero-day attacks.

[2] Nandiraju, S. K. K., et al., "Enhancing Cybersecurity: Zero-Day Attack Detection in Network Traffic with Deep Learning," Asian Journal of Research in Computer Science, 2025. Uses a Convolutional Neural Network (CNN) on the UNSW-NB15 dataset for accurate zero-day detection.

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[4] Nhlapo, S. J., & Nkongolo Waa Nkongolo, "Zero-Day Attack and Ransomware Detection Using ML Classifiers," arXiv, 2024. Evaluates machine learning models such as Random Forest and XGBoost for ransomware and zero-day detection.

[5] Yang Guo, "A Review of Machine Learning-Based Zero-Day Attack Detection: Challenges and Future Directions," Computer Communications, 2023. A comprehensive survey of machine-learning techniques used to identify zero-day attacks.